The Law of Cosines

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Vectors and the dot product

A vector describes a displacement, with both a length and a direction. For vectors $u = (u_1, u_2)$ and $v = (v_1, v_2)$ in the plane, their *dot product* is the real number

$$u \cdot v = u_1 v_1 + u_2 v_2.$$

The length, or *norm*, of u is

$$\|u\| = \sqrt{u_1^2 + u_2^2}, \qquad \text{so} \qquad \|u\|^2 = u \cdot u.$$

The same definitions work in three dimensions by including the third coordinate, and in higher dimensions by summing over all coordinates. The coordinate definition gives

$$u \cdot v = v \cdot u, \qquad u \cdot (v + w) = u \cdot v + u \cdot w, \qquad u \cdot (tv) = t(u \cdot v)$$

for any real number t.

Dot-product theorem. *If u and v are nonzero vectors and $\theta \in [0, \pi]$ is the angle between them, then*

$$\boxed{u \cdot v = \|u\| \, \|v\| \cos\theta.}$$

We state this theorem without proof. The dot product is a scalar, so no vector norm is needed on its left side. In particular, it is negative when the angle is obtuse.

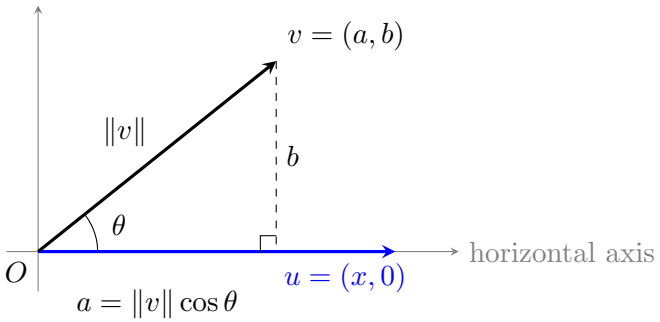
Seeing the formula when one vector is horizontal

Suppose $u = (x, 0)$ with $x > 0$, and write $v = (a, b) \neq (0, 0)$. The horizontal component of v is its signed projection onto the horizontal axis:

$$a = \|v\| \cos\theta.$$

For an acute angle, this is just the adjacent side divided by the hypotenuse definition of cosine in the right triangle below. Thus the dot-product formula is easy to see in this case:

$$u \cdot v = xa + 0b = x \, \|v\| \cos\theta = \|u\| \, \|v\| \cos\theta.$$



The diagram shows an acute angle with $b > 0$. The signed-projection identity also holds for right and obtuse angles: when θ is obtuse, both a and $\cos\theta$ are negative. If $x < 0$, the vector u points left, so instead $a = -\|v\| \cos\theta$. Since $\|u\| = |x| = -x$, we again obtain $u \cdot v = xa = \|u\| \, \|v\| \cos\theta$. The case $x = 0$ gives the zero vector, for which the angle is undefined.

The law of cosines

In a triangle ABC, let

$$a = |BC|, \qquad b = |AC|, \qquad c = |AB|, \qquad \theta = \angle BAC.$$

Thus θ is the angle between the sides of lengths b and c, and a is the length of the opposite side. The *law of cosines* states that

$$\boxed{a^2 = b^2 + c^2 - 2bc\cos\theta.}$$

Proof. Set $u = \overrightarrow{AB}$ and $v = \overrightarrow{AC}$. Then $\overrightarrow{BC} = v - u$, so $a = \|v - u\|$, $b = \|v\|$, and $c = \|u\|$. Expanding the dot product and applying the theorem gives

$$\begin{aligned} a^2 &= (v - u) \cdot (v - u) \\ &= v \cdot v + u \cdot u - 2u \cdot v \\ &= \|v\|^2 + \|u\|^2 - 2 \|u\| \, \|v\| \cos\theta \\ &= b^2 + c^2 - 2bc\cos\theta. \end{aligned}$$

□

When $\theta = 90°$, its cosine is zero, and the law of cosines reduces to the Pythagorean theorem.

The length of a sum

For two nonzero vectors with angle θ between them,

$$\boxed{\|u + v\|^2 = \|u\|^2 + \|v\|^2 + 2 \|u\| \, \|v\| \cos\theta.}$$

Proof. Using the dot-product identities and the theorem,

$$\begin{aligned} \|u + v\|^2 &= (u + v) \cdot (u + v) \\ &= u \cdot u + 2u \cdot v + v \cdot v \\ &= \|u\|^2 + \|v\|^2 + 2 \|u\| \, \|v\| \cos\theta. \end{aligned}$$

□

The minus sign in the law of cosines comes from the difference $v - u$; the plus sign here comes from the sum $u + v$. If either vector is zero, the angle is undefined, but the angle-free identity $\|u + v\|^2 = \|u\|^2 + \|v\|^2 + 2u \cdot v$ still holds.

Exercises

1. For $u = (1, 2)$ and $v = (3, 0)$, compute $u \cdot v$, $\|u\|$, and $\|v\|$.

2. Two sides of a triangle have lengths 3 and 4, and their included angle is $90°$. Find the opposite side.

3. Two sides of a triangle each have length 2, and their included angle is $60°$. Use the law of cosines to find the opposite side.

4. Suppose $\|u\| = 2$, $\|v\| = 3$, and the angle between them is $60°$. Find $\|u + v\|^2$.

5. A triangle has side lengths 3, 5, and 7. Find the cosine of the angle opposite the side of length 7. Is this angle acute, right, or obtuse?

A Answers to the exercises

1. The coordinate definitions give

$$u \cdot v = 1(3) + 2(0) = 3, \qquad \|u\| = \sqrt{1^2 + 2^2} = \sqrt{5}, \qquad \|v\| = 3.$$

2. If a is the opposite side, then $a^2 = 3^2 + 4^2 - 2(3)(4)\cos 90^\circ = 25$. Hence $a = 5$.

3. Using $\cos 60^\circ = \frac{1}{2}$, $a^2 = 2^2 + 2^2 - 2(2)(2)(\frac{1}{2}) = 4$. Hence $a = 2$.

4. The sum formula gives

$$\|u + v\|^2 = 2^2 + 3^2 + 2(2)(3)(\tfrac{1}{2}) = 19.$$

5. Let θ be the angle opposite the side of length 7. The law of cosines gives

$$\cos\theta = \frac{3^2 + 5^2 - 7^2}{2(3)(5)} = \frac{-15}{30} = -\frac{1}{2}.$$

   Thus $\theta = 120^\circ$, which is obtuse.